

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 6: Microwave Landing System (MLS) & GLS/SBAS

Syllabus Source: CAE Oxford (pp. 217-252) | Pass Target: $\geq 90\%$ AviationExam

1. MICROWAVE LANDING SYSTEM (MLS)

MLS was developed to overcome ILS frequency saturation and multipath limitations. It operates in the SHF (C-band) from 5030 to 5091 MHz (200 channels):

- Time-Referenced Scanning Beam (TRSB): A narrow beam sweeps back-and-forth across the coverage sector (TO and FRO scan). The time interval between the TO and FRO beam pulses received by the aircraft directly indicates azimuth and elevation angles!
- Operational Capabilities: Azimuth coverage: $\pm 40^\circ$ (up to $\pm 60^\circ$); Elevation: 0.9° to 15° ; Range: 20 NM. Enables CURVED AND SEGMENTED APPROACHES for noise abatement and terrain clearance.

2. SATELLITE-BASED LANDING SYSTEMS (GLS / GBAS)

The Ground-Based Augmentation System (GBAS) provides differential GPS corrections enabling GLS (GBAS Landing System) Cat I, II, and III precision approaches:

- How GLS Works: 4 reference GPS receivers on the airport measure satellite pseudo-range errors. A ground station calculates corrections and broadcasts them along with Final Approach Segment Data Blocks (FASB) over a VHF Data Broadcast (VDB, 108-118 MHz).
- Pilot Interface: The pilot selects a 5-digit channel number on the nav control panel. The cockpit display is identical to ILS, but with pinpoint differential accuracy and zero beam scalloping!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Precision Approach Categories &

EASA Official Approach Minimums: - Category I (CAT I): Decision Height (DH) ≥ 200 ft. RVR ≥ 550 meters. - Category II (CAT II): Decision Height (DH) between 100 ft and 200 ft. RVR ≥ 300 meters. - Category IIIA: Decision Height (DH) < 100 ft (or no DH). RVR ≥ 175 meters. - Category IIIB: Decision Height (DH) < 50 ft (or no DH). RVR ≥ 75 meters. - Category IIIC: Zero Decision Height and Zero RVR (Blind autoland and roll-out).

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In which frequency band does the Microwave Landing System (MLS) operate?

- [A]: UHF
- [B]: SHF (C-Band: 5030 MHz to 5091 MHz)
- [C]: VHF
- [D]: EHF

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. MLS operates in the SHF C-band, rendering it immune to low-frequency site reflections and FM radio interference.

Q2: What is the primary operational advantage of MLS over conventional ILS?

- [A]: Lower cost
- [B]: Wide-angle proportional coverage permitting CURVED AND SEGMENTED APPROACH PATHS with variable glide slope angles
- [C]: Operates without electricity
- [D]: Long-range en route guidance

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. MLS provides wide azimuth ($\pm 40^\circ$) and elevation coverage, allowing aircraft to fly curved noise-abatement profiles down to touchdown.

Q3: In a GBAS Landing System (GLS), how are the ground differential corrections transmitted to the aircraft?

- [A]: Via satellite phone
- [B]: Via a VHF DATA BROADCAST (VDB) in the 108.000 to 117.975 MHz band
- [C]: Via HF sky wave
- [D]: Via primary radar

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The GBAS ground station broadcasts corrections via an omnidirectional VHF Data Broadcast antenna at the airport.